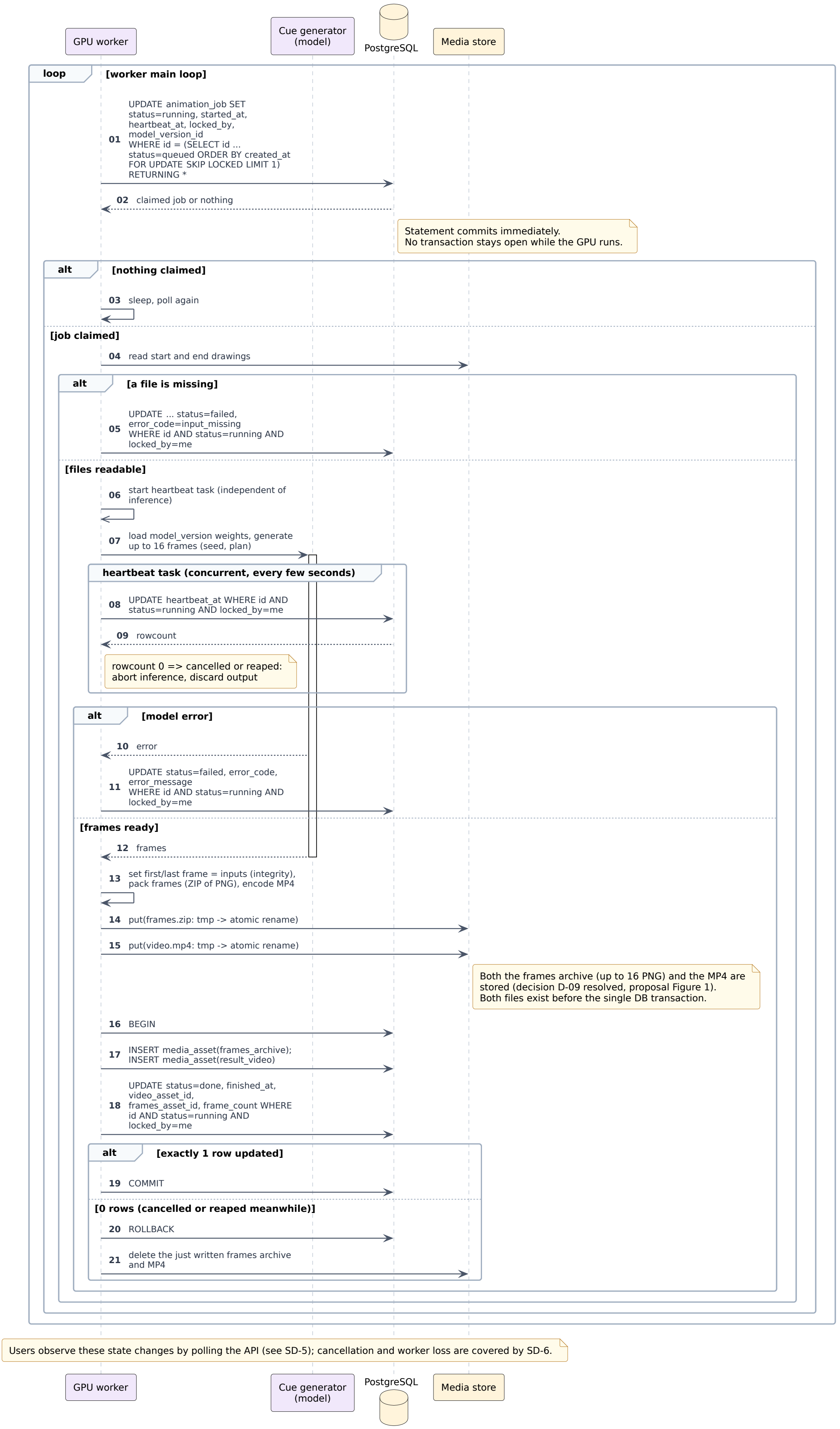SD-4b Animation generation — asynchronous execution by the GPU worker (FR-06, FR-07, FR-10, NFR-02, NFR-03)